Online Structured Sparsity-Based Moving-Object Detection From Satellite Videos

Junpeng Zhang, *Student Member, IEEE*, Xiuping Jia, *Senior Member, IEEE*, Jiankun Hu, *Senior Member, IEEE*, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— Inspired by the recent developments in computer vision, low-rank and structured sparse matrix decomposition can be potentially be used for extract moving objects in satellite videos. This set of approaches seeks for rank minimization on the background that typically requires batch-based optimization over a sequence of frames, which causes delays in processing and limits their applications. To remedy this delay, we propose an online low-rank and structured sparse decomposition (O-LSD). O-LSD reformulates the batch-based low-rank matrix decomposition with the structured sparse penalty to its equivalent framewise separable counterpart, which then defines a stochastic optimization problem for online subspace basis estimation. In order to promote online processing, O-LSD conducts the foreground and background separations and the subspace basis update alternatingly for every frame in a video. We also show the convergence of O-LSD theoretically. Experimental results on two satellite videos demonstrate the performance of O-LSD in terms of accuracy, and the time consumption is comparable with the batch-based approaches with significantly reduced delay in processing.**

***Index Terms*— Background subtraction, moving-object detection (MOD), online robust principal component analysis (OR-PCA), satellite video processing, structured sparsity-inducing norm.**

I. Introduction

OBJECT detection on high-resolution aerial images has been actively investigated in recent years [1], [2]. Inspired by the state-of-the-art deep learning methods, such as regional convolution neural network (R-CNN) [3], Fast R-CNN [4], Faster R-CNN [5], You Only Look Once (YOLO) [6], and single-shot multibox detector (SSD) [7], object detection performance on these images has been improved significantly [8]–[11]. These approaches are mainly exploring the spectral (or color) and spatial (texture or context) information on objects of interest, and they detect objects of interest image by image, as no temporal information is available on those images. Recently, with satellite videos captured by Jilin-1 [12] and Skybox [13], dense temporal information becomes available, which benefits moving-object detection (MOD) from space. Target tracking becomes possible and can then be conducted for various applications [14]–[17].

Detecting moving objects in a video is achieved by separating the temporal varying foreground, which is associated with the moving objects, and the background that lays in a low-dimensional subspace from a video [18]–[20]. Given the moving objects account for a limited number of pixels in the foreground, the foreground is assumed sparse. Robust principal component analysis (RPCA), as one fundamental method in foreground extraction, defines a low-rank matrix decomposition problem with a sparse penalty [21], which is solved by principal component pursuit (RPCA-PCP) [21]–[23] and fast low-rank approximation (GoDec) [24]. Based on the duality between the sparsity and the Laplace distributions, probabilistic robust matrix factorization (PRMF) provides a probabilistic interpretation to RPCA by combining the Laplace error and the Gaussian prior [25].

As a moving object is commonly a set of neighboring pixels, spatial prior to the foreground is considered in low-rank matrix decomposition to improve MOD performance. Total variation (TV) regularization is introduced to enforce smoothness on the foreground in the matrix decomposition [26]. DEtecting Contiguous Outliers in the LOw-rank Representation (DECOLOR) constrains the edges of moving objects to be contiguous, and then, the first-order Markov random field (MRF) is integrated into low-rank matrix decomposition [27]. Another possible spatial prior to the foreground is the sparsity over groups of spatial neighboring pixels on the foreground other than pixelwise sparse, which is measured by structured sparsity-inducing norm [28]. Low-rank and structured sparse decomposition (LSD) obeys this prior and penalizes the low-rank matrix decomposition by the structured sparsity-inducing norm of the foreground [29]. As MOD in satellite video is more sensitive to random noises, integrating spatial prior should improve the quality of the estimated foreground, thus the moving-object detection performance. By integrating structured sparsity, LSD presents boosted MOD performance in satellite videos [30]. DECOLOR, however, has a limited improvement in MOD performance for satellite videos, as the introduced MRF constraint tends to merge neighboring targets when the distance between them is too small.

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Junpeng Zhang, Xiuping Jia, and Jiankun Hu are with the School of Engineering and Information Technology, University of New South Wales, Canberra, ACT 2612, Australia (e-mail: junpeng.zhang@student.unsw.edu.au; x.jia@adfa.edu.au; j.hu@adfa.edu.au).

Jocelyn Chanussot is with University of Grenoble Alpes, Inria, CNRS, Grenoble INP, LJK, 38000 Grenoble, France (e-mail: jocelyn.chanussot@grenoble-inp.fr).

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TABLE I
COMPARISON ON ONLINE LOW-RANK MATRIX DECOMPOSITION ALGORITHMS FOR MOD

Method	Objective Function	Constraints	Spatial Prior	Optimization Scheme	Proven Convergence
OPRMF [25]	$\min \lambda \|\mathbf{D}-\mathbf{LR}\|_1 + \frac{\lambda_1}{2}\|\mathbf{L}\|_F^2 + \frac{\lambda_2}{2}\|\mathbf{R}\|_F^2$	$\mathbf{L}_{ij}\|\lambda_1 \sim N(\mathbf{L}_{ij}\|\mathbf{0}, \lambda_1^{-1})$, $\mathbf{R}_{ij}\|\lambda_2 \sim N(\mathbf{R}_{ij}\|\mathbf{0}, \lambda_2^{-1})$	-	Online Expectation Maximization	No
GRASTA [31]	$\min \|\mathbf{s}\|_1$	$\mathbf{d} = \mathbf{Lr} + \mathbf{s}$, $\mathbf{L}^T\mathbf{L} = \mathbf{I}$	-	Incremental Gradient Descent Method on Grassmannian Manifold	No
OR-PCA [32]	$\min \frac{1}{2}\|\mathbf{D}-\mathbf{LR}-\mathbf{S}\|_F^2 + \frac{\lambda_1}{2}\|\mathbf{L}\|_F^2 + \frac{\lambda_1}{2}\|\mathbf{R}\|_F^2 + \lambda_2\|\mathbf{S}\|_1$	-	-	Stochastic Optimization	**Yes**
COROLA [33]	$\min \frac{1}{2}\|\mathcal{P}_{S^\perp}(\mathbf{D}-\mathbf{LR})\|_F^2 + \frac{\alpha}{2}\|\mathcal{P}_{S^\perp}(\mathbf{L})\|_F^2 + \frac{\alpha}{2}\|\mathcal{P}_{S^\perp}(\mathbf{R})\|_F^2 + \beta\|\mathbf{S}\|_1 + \gamma\|Avec(\mathbf{S})\|$	$\mathbf{S}_{ij} \in \{0,1\}$	Edge Contiguousness	Stochastic Optimization	**Yes**
GOSUS [34]	$\min \sum_{i=1}^{l} \mu_i \|\mathbf{Gs}\|_2 + \frac{\lambda}{2}\|\mathbf{Lr}+\mathbf{s}-\mathbf{d}\|_2^2$	$\mathbf{L}^T\mathbf{L} = \mathbf{I}$	**Structured Sparsity of Foreground**	Incremental Gradient Descent Method on Grassmannian Manifold	No
Proposed O-LSD	$\min \frac{1}{2}\|\mathbf{D}-\mathbf{LR}-\mathbf{S}\|_F^2 + \frac{\lambda_1}{2}\|\mathbf{L}\|_F^2 + \frac{\lambda_1}{2}\|\mathbf{R}\|_F^2 + \lambda_2 \sum_{\mathbf{s}\in\mathbf{S}} \|\mathbf{s}\|_{\ell_1/\ell_\infty}$	-	**Structured Sparsity of Foreground**	Stochastic Optimization	**Yes**

∗There exist a variety of low-rank decomposition algorithm in the literature, however, we select the most related works here. Interested readers may refer to [18], [19], [20], [35] for more comprehensive reviews in low-rank matrix decomposition and their applications in video processing.

Regardless of the detection performance of the algorithms mentioned earlier, a pitfall of them is that their solutions are based on optimization in a batch manner. These approaches use singular value decomposition (SVD) for low-rank background estimation, which couples all the samples in each iteration of the optimization. The detection results are not available until the optimization terminates, which results in delays in processing. Another shortcoming of batch-based approaches is the difficulty in handling a video with an incremental length. Both issues of the batch-based algorithms limit their application in various online systems.

In order to reduce the delay and to make MOD adaptable to videos of incremental lengths, the online method is expected to sequentially estimate the foreground and background for each new incoming frame. By low-rank matrix decomposition, the estimated principal component basis vectors represent a subspace, where the background lays. This subspace can be identified by a point on the Grassmannian manifold, and the incremental gradient descent method on Grassmannian Manifold is employed for online subspace tracking or updating [31]–[33]. For MOD, the Grassmannian robust adaptive subspace tracking algorithm (GRASTA) [34] and the Grassmannian online subspace updates with structured-sparsity (GOSUS) [35] are developed for online low-rank matrix decomposition with the pixelwise sparse penalty and the structured sparse penalty, respectively. These sets of approaches, however, provide no theoretical guarantee on their convergence, and their performance is heavily sensitive to the selection of the learning rate.

Another possibility for online low-rank matrix decomposition is the increasingly common matrix factorization approximation of nuclear norm, where the rank minimization is replaced by the sum of square penalties of its factorization [36]–[40]. With this reformulation, iterative optimization scheme is then developed based on stochastic optimization for solving low-rank decomposition problems online. Online robust principal component analysis (OR-PCA) solves the online low-rank matrix decomposition problem with pixelwise sparsity penalty, which, more importantly, proves that the iterative optimization algorithm converges to the global optimum of the original RPCA approach [37]. Utilizing first-order MRF, spatial prior to contiguous edges is also integrated with this reformulation for online low-rank matrix decomposition [39].

It has been observed that by introducing the structured sparse penalty, LSD boosts the MOD performance in satellite videos [30]. While GOSUS combines online low-dimensional subspace tracking with structured sparsity, no theoretical guarantee on its convergence is provided, as presented in Table I. In order to fill this gap, we present an online low-rank matrix decomposition approach with the structured sparse penalty, named online low-rank and structured sparse decomposition (O-LSD), which not only combines the structured sparsity penalty but also provides theoretically guaranteed convergence. We follow the matrix factorization approximation of nuclear norm for online learning in [32], [33], [40], and [41] and decompose the background matrix to a set of background frames that are reconstructed by the estimated subspace basis and their associated coefficients. To promote online processing, the proposed O-LSD algorithm is composed of two building blocks. For each frame, its corresponding foreground and background frames are reconstructed by the current subspace basis, and then, the subspace basis is updated for this new input. This procedure defines a stochastic optimization problem, and we show that the O-LSD algorithm converges almost surely. Existing convergence analysis in [32] and [41]–[43] is built on necessary and sufficient conditions on the unique solution in sparse encoding. In O-LSD, no such conditions exist for structured sparsity encoding to the best of our knowledge, and we show the convergence

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of O-LSD based on the boundedness of the subgradients in structured sparsity encoding; then, a set of related properties of O-LSD is demonstrated. Experimental evaluations and analyses were performed on a satellite video data set with two videos, where we compared our algorithm with five state-of-the-art algorithms.

In summary, the main contributions of this article are fourfold.

1) We propose an O-LSD for MOD in satellite videos by reformulating batch-based LSD using the matrix factorization approximation of nuclear norm.
2) To solve the new reformulated optimization problem, two iterating steps are designed and developed. For each frame, the corresponding foreground and background frames are first reconstructed by the current subspace basis, and then, the subspace basis is updated by the given frame.
3) On the contrary to most current online algorithms, we show that O-LSD can converge without meeting the conditions for the unique solution of structured sparsity encoding. Due to the lack of these conditions, the solution of O-LSD can converge to neither a stationary point nor the global optimum of its batch-based counterpart LSD. This finding and its corresponding proof are useful beyond the scope of this article.
4) Due to the better convergence characteristics of O-LSD, it can further reduce its processing delay with negligible effects on the detection performance, by downsampling in the temporal domain.

The remainder of this article is organized as follows. The proposed O-LSD is presented in Section II, where its convergence analysis is provided in Section II-E. The experimental parameter settings and performance comparison against state-of-the-art approaches are presented in Section III. Finally, conclusions and suggestions for future research are given in Section IV.

II. Proposed Method

A. Matrix Factorization to LSD

LSD seeks a low-rank matrix decomposition of an observation matrix, which, at the same time, imposes the structured sparse penalty on the foreground. LSD defines a low-rank matrix decomposition approach for MOD, where the structure of both background and foreground is taken into consideration. In a satellite video, the background lays in a low-dimensional subspace, whereas the moving objects of interest are contained in the temporally changing foreground. Therefore, detecting moving objects in a satellite video can be achieved by separating the foreground and background. Since the moving objects are groups of neighboring pixels instead of isolate pixels, LSD also imposes the structured sparse regularization on the foreground.

Given a fixed-length sequence from n video frames and each frame contains p pixels, LSD decomposes its corresponding matrix $\mathbf{D} \in \mathbb{R}^{p\times n}$ to a low-rank background matrix $\mathbf{B} \in \mathbb{R}^{p\times n}$ plus a structured sparse foreground matrix $\mathbf{S} \in \mathbb{R}^{p\times n}$ and defines a batch-based optimization problems as

$$(\mathbf{B}^*, \mathbf{S}^*) = \arg\min_{\mathbf{B},\mathbf{S}} \|\mathbf{B}\|_* + \lambda\Omega(\mathbf{S}) \quad \text{s.t. } \mathbf{D} = \mathbf{B} + \mathbf{S} \tag{1}$$

where $\Omega(\mathbf{S})$ refers to the structured sparsity-inducing norm of $\mathbf{S}$ and λ is a scalar that assigns the weight of structured sparsity. The structured sparsity-inducing norm [28], [43], [44] indicates the sparsity over groups of neighboring pixels as

$$\Omega(\mathbf{S}) = \sum_{\mathbf{s}\in\mathbf{S}} \|\mathbf{s}\|_{\ell_1/\ell_\infty} = \sum_{\mathbf{s}\in\mathbf{S}} \sum_{g\in\mathcal{G}} \eta_g \left\|\mathbf{s}_{|g}\right\|_\infty \tag{2}$$

where $\mathcal{G}$ defines the set of groups of neighboring pixels and $\mathbf{s}_{|g} \in \mathbb{R}^p$ is a sparse vector with nonzero elements at the indices represented in a group $g \in \mathcal{G}$. η_g denotes the weight for a group of the pixels. In this article, we assume that each group contributes equally and assign $\eta_g = 1.0\ \forall g \in \mathcal{G}$. In LSD, no temporal prior or constraints on the foreground are considered; thus, the structured sparse penalty over a sequence of frames is framewise separable. In this article, the groups of spatially related pixels $\mathcal{G}$ are constructed by 3×3 grid scanning over the foreground, as the moving targets in satellite videos are usually in small scales.

Inspired by [37], the equality constraint in (1) is removed, and we obtain a reformulated optimization problem

$$(\mathbf{B}^*, \mathbf{S}^*) = \arg\min_{\mathbf{B},\mathbf{S}} \lambda_1\|\mathbf{B}\|_* + \lambda_2 \sum_{\mathbf{s}\in\mathbf{S}} \|\mathbf{s}\|_{\ell_1/\ell_\infty} + \frac{1}{2}\|\mathbf{D} - \mathbf{B} - \mathbf{S}\|_F^2 \tag{3}$$

in which $\lambda_1 > 0$ and $\lambda_2 > 0$ are the corresponding weights for the low-rank penalty and the structured sparsity penalty.

Guided by the trending reformulation by matrix factorization in [32], [35], and [40], we replace the low-rank term $\|\mathbf{B}\|_*$ in (3) by its approximation, which makes use of Lemma 1.

Lemma II.1: *Given that* $\mathbf{B}$ *is factorized as* $\mathbf{B} = \mathbf{LR}$, $L \in \mathbb{R}^{p*r}$, *and* $\mathbf{R} \in \mathbb{R}^{r*n}$, *the nuclear norm of* $\mathbf{B}$ *is upper bounded by the sum of Frobenius norms of* $\mathbf{L}$ *and* $\mathbf{R}$, *as*

$$\|\mathbf{B}\|_* = \inf_{\mathbf{L}\in\mathbb{R}^{p\times r},\mathbf{R}\in\mathbb{R}^{r\times n}} \left\{\frac{1}{2}\|\mathbf{L}\|_F^2 + \frac{1}{2}\|\mathbf{R}\|_F^2 : \mathbf{B} = \mathbf{LR}\right\}. \tag{4}$$

When $r > \text{Rank}(\mathbf{B})$, *the jointly nonconvex quadratic optimization problem (4) is equivalent to minimize the nuclear norm of* $\mathbf{B}$.[1]

By substituting $\|\mathbf{B}\|_*$ with its factorized approximation, we rewrite the optimization problem in (3) to

$$(\mathbf{L}^*, \mathbf{R}^*, \mathbf{S}^*) = \arg\min_{\mathbf{L},\mathbf{R},\mathbf{S}} \frac{1}{2}\|\mathbf{D} - \mathbf{LR} - \mathbf{S}\|_F^2 + \frac{\lambda_1}{2}\|\mathbf{L}\|_F^2 + \frac{\lambda_1}{2}\|\mathbf{R}\|_F^2 + \lambda_2 \sum_{\mathbf{s}\in\mathbf{S}} \|\mathbf{s}\|_{\ell_1/\ell_\infty} \tag{5}$$

where $\mathbf{L} \in \mathbb{R}^{p*r}$ is considered as the subspace basis of the background matrix $\mathbf{B}$, and $\mathbf{R} \in \mathbb{R}^{r*n}$ is the coefficients to reconstruct $\mathbf{B}$ with given $\mathbf{L}$. r is the estimated dimension of the subspace that the background frames lay in.

With a pair of estimated $\mathbf{L}$ and $\mathbf{R}$, each column vector in $\mathbf{R}$ corresponds to an estimated background frame in $\mathbf{B}$.

[1] Please refer to [36] for detailed proof.

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Let $\mathbf{d}_t$, $\mathbf{r}_t$, and $\mathbf{s}_t$ refer to the tth column of $\mathbf{D}$, $\mathbf{R}$, and $\mathbf{S}$, respectively, and this optimization problem in (5) is equivalent to minimizing an empirical cost function

$$f_n(\mathbf{L}) = \frac{1}{n}\sum_{i=1}^{n} \ell(\mathbf{D}_i, \mathbf{L}) + \frac{\lambda_1}{2n}\|\mathbf{L}\|_F^2 \tag{6}$$

in which $\ell(\mathbf{d}_i, \mathbf{L})$ is the reconstruction cost evaluated with fixed $\mathbf{L}$ by

$$\ell(\mathbf{d}, \mathbf{L}) = \min_{\mathbf{r},\mathbf{s}} \hat{\ell}(\mathbf{d}, \mathbf{L}, \mathbf{r}, \mathbf{s})$$
$$\hat{\ell}(\mathbf{d}, \mathbf{L}, \mathbf{r}, \mathbf{s}) = \frac{1}{2}\|\mathbf{d} - \mathbf{L}\mathbf{r} - \mathbf{s}\|_2^2 + \frac{\lambda_1}{2}\|\mathbf{r}\|_2^2 + \lambda_2\|\mathbf{s}\|_{\ell_1/\ell_\infty}. \tag{7}$$

By dividing (5) with the number of samples, the empirical costs with different number of samples become comparable, which is critical since n is gradually increasing in online algorithm. Minimization of the empirical cost function in (6) associates the sum of reconstruction costs $\ell(\cdot, \mathbf{L})$, where, for each frame, $(\mathbf{r}, \mathbf{s})$ is optimized with the optimization target $\mathbf{L}$ as a parameter, whose formulation fits to the max–min optimization problem.

B. O-LSD

Through the above-mentioned reformulation, it is still impossible to update $\mathbf{L}$ without reestimating all pairs of $(\mathbf{r}, \mathbf{s})$, which obstructs processing in an online fashion. In order to promote online processing, we propose an online algorithm, named O-LSD, where the foreground and background separations and subspace basis update are sequentially conducted for each frame.

O-LSD is an online algorithm that processes an input frame at each time instance in an online manner. At each time instance t, we have obtained $\mathbf{L}_{t-1}$ estimated from previous time instance $t-1$. The foreground frame and background frame are separated by solving the following optimization problem:

$$(\mathbf{r}_t^*, \mathbf{s}_t^*) = \arg\min_{\mathbf{r}_t,\mathbf{s}_t} \hat{\ell}(\mathbf{d}_t, \mathbf{L}_{t-1}, \mathbf{r}_t, \mathbf{s}_t). \tag{8}$$

We term this procedure as foreground and background separations, which is detailed in Section II-C.

Then, subspace basis updating is performed with all pair of $\mathbf{r}_i$ and $\mathbf{s}_i$, $i \in \{1, \ldots, t-1\}$.

Directly minimizing the empirical cost function defined in (6) requires reestimations on all pairs of $\mathbf{r}_i$ and $\mathbf{s}_i$. Alternatively, the subspace basis $\mathbf{L}_t$ is updated by minimizing a surrogate function of the empirical cost function $g_t(\mathbf{L}_t)$, which provides an upper bound for $f_t(\mathbf{L}_t)$ so that $g_t(\mathbf{L}_t) > f_t(\mathbf{L}_t)$. By introducing the surrogate function, no changes are required on the estimated pairs of $\mathbf{r}_i$ and $\mathbf{s}_i$, $i \in \{1, \ldots, t-1\}$. We define the surrogate function as

$$\begin{aligned} g_t(\mathbf{L}_t) &= \frac{1}{t}\sum_{t=1}^{\infty} \hat{\ell}(\mathbf{d}_t, \mathbf{L}_t, \mathbf{r}_t, \mathbf{s}_t) + \frac{\lambda_1}{2t}\|\mathbf{L}_t\|_F^2 \\ &= \frac{1}{t}\sum_{i=1}^{t}\left(\frac{1}{2}\|\mathbf{d}_i - \mathbf{L}_t\mathbf{r}_i - \mathbf{s}_i\|_2^2 + \frac{\lambda_1}{2}\|\mathbf{r}_i\|_2^2 \right. \\ &\qquad \left. + \lambda_2\|\mathbf{s}_i\|_{\ell_1/\ell_\infty}\right) + \frac{\lambda_1}{2t}\|\mathbf{L}_t\|_F^2. \end{aligned} \tag{9}$$

Algorithm 1 Proposed O-LSD Algorithm for MOD

Input: $\mathbf{d}_t \in \mathbb{R}^p$, $\mathbf{L}_{t-1} \in \mathbb{R}^{p\times r}$, $\mathbf{A}_{t-1}$ and $\mathbf{B}_{t-1}$
Output: $\mathbf{b}_t$, $\mathbf{r}_t$, $\mathbf{s}_t$ and $\mathbf{L}_t$

1: Separate the foreground and background:

$$(\mathbf{r}_t^*, \mathbf{s}_t^*) = \arg\min_{\mathbf{r}_t,\mathbf{s}_t} \frac{1}{2}\|\mathbf{d}_t - \mathbf{L}_{t-1}\mathbf{r}_t - \mathbf{s}_t\|_2^2 + \frac{\lambda_1}{2}\|\mathbf{r}_t\|_2^2 + \lambda_2\|\mathbf{s}_t\|_{\ell_1/\ell_\infty},$$

which is solved by Algorithm 2.

2: Compute the background frame: $\mathbf{b}_t = \mathbf{L}_{t-1}\mathbf{r}_t$.

3: Update the accumulation matrices $\mathbf{A}_t$ and $\mathbf{B}_t$ by Eq. 17.

4: Update the subspace basis $\mathbf{L}_t$:

$$\mathbf{L}_t^* = \arg\min_{\mathbf{L}_t} \operatorname{Tr}(\mathbf{L}_t^T(\lambda_1\mathbf{I} + \mathbf{A}_t)\mathbf{L}_t) - 2\operatorname{Tr}(\mathbf{L}_t^T\mathbf{B}_t),$$

whose solution is presented in Algorithm 3.

5: **return** $\mathbf{b}_t$, $\mathbf{r}_t$, $\mathbf{s}_t$ and $\mathbf{L}_t$.

The minimization of $g_t(\mathbf{L}_t)$ with respect to $\mathbf{L}_t$ is termed as the subspace basis update, which is then explained in Section II-D. The entire O-LSD algorithm is summarized in Algorithm 1.

C. Foreground and Background Separations

Foreground and background separations obtain a pair of $\mathbf{r}_t^*$ and $\mathbf{s}_t^*$ by solving the optimization problem defined in (8), where $\mathbf{L}_{t-1}$ is provided by the previous time instance $t-1$. As $[\mathbf{L}\ \mathbf{I}]^T[\mathbf{L}\ \mathbf{I}]$ is always positive semidefinite, the objective function of (8) is convex with respect to $(\mathbf{r}_t, \mathbf{s}_t)$. The pair of variable r_t and s_t are independent of r_{t-1} and s_{t-1}, and we solve them by a block coordinate descent (BCD) method [46], where the variables are estimated by solving subproblems with each variable fixed in turn.

By fixing $\mathbf{s}_t$, $\mathbf{r}_t^*$ is obtained by solving

$$\mathbf{r}_t^* = \arg\min_{r_t} \frac{1}{2}\|\mathbf{d}_t - \mathbf{L}_{t-1}\mathbf{r}_t - \mathbf{s}_t\|_2^2 + \frac{\lambda_1}{2}\|\mathbf{r}_t\|_2^2 \tag{10}$$

which constructs a least-square problem, and its closed-form solution is given by

$$\mathbf{r}_t^* = (\mathbf{L}_{t-1}^T\mathbf{L}_{t-1} + \lambda_1 I)^{-1}(\mathbf{d}_t - \mathbf{s}_t). \tag{11}$$

Using fixed $\mathbf{r}_t$, let $\mathbf{u} = \mathbf{d}_t - \mathbf{L}_{t-1}\mathbf{r}_t$, the subproblem for estimating the structured sparsity $\mathbf{s}_t$ is defined as

$$\mathbf{s}_t^* = \arg\min_{\mathbf{s}_t} \frac{1}{2}\|\mathbf{u} - \mathbf{s}_t\|_2^2 + \lambda_2\|\mathbf{s}_t\|_{\ell_1/\ell_\infty} \tag{12}$$

whose solution is obtained by its dual problem that define a quadratic min-cost flow problem [41], [47], [48] as

$$\xi^* = \arg\min_{\xi} \frac{1}{2}\left\|\mathbf{u} - \sum_{g\in\mathcal{G}}\xi^g\right\|_2^2 \quad \text{s.t. } \forall g \in \mathcal{G},\quad \|\xi^g\|_1 \le \lambda_2 \ \text{ and } \ \xi_j^g = 0 \text{ if } j \notin g \tag{13}$$

where $\xi^g \in \mathbb{R}^p\ \forall g \in \mathcal{G}$ denotes the corresponding dual variables for the group of variables in g, and ξ is the set

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Algorithm 2 BCD Method for Foreground and Background Separations

Input: $\mathbf{L}_{t-1} \in \mathbb{R}^{p*r}$, $\lambda_1 > 0$ and $\lambda_2 > 0$
Output: $\mathbf{r}_t$ and $\mathbf{s}_t$
1: $\mathbf{r}_t = \mathbf{0}$, $\mathbf{s}_t = \mathbf{0}$.
2: **while** not converged **do**
3: Estimate $\mathbf{r}_t$:

$$\mathbf{r}_t = (\mathbf{L}_{t-1}^T \mathbf{L}_{t-1} + \lambda_1 I)^{-1}(\mathbf{d}_t - \mathbf{s}_t).$$

4: Estimate $\mathbf{s}_t$ by solving its dual problem on ξ defined in Eq. 13, then

$$\mathbf{s}_t = \mathbf{d}_t - \mathbf{L}_{t-1}\mathbf{r}_t - \sum_{g\in\mathcal{G}} \xi^{*g}.$$

5: Check the convergence using Eq. 15.
6: **end while**
7: **return** $\mathbf{r}_t$ and $\mathbf{s}_t$

of all ξ^g $\forall g \in \mathcal{G}$. The primal solution $\mathbf{s}_t$ is then obtained by

$$\mathbf{s}_t^* = \mathbf{u} - \sum_{g\in\mathcal{G}} \xi^{*g}. \tag{14}$$

The iteration of alternatingly estimation of $\mathbf{r}_t$ and $\mathbf{s}_t$ continues until the stop criterion is reached

$$\frac{\max\left\{\left\|\mathbf{r}_t' - \mathbf{r}_t''\right\|_2, \left\|\mathbf{s}_t' - \mathbf{s}_t''\right\|_2\right\}}{p} \le \tau \tag{15}$$

where $(\mathbf{r}_t', \mathbf{s}_t')$ and $(\mathbf{r}_t'', \mathbf{s}_t'')$ are the two pairs of estimation solutions at two consecutive iterations. Similar to [33] and [35], the stop criterion is set as $\tau = 1.0e-5$ in this article. The BCD algorithm for foreground and background separations is summarized in Algorithm 2.

D. Subspace Basis Update

After estimating $(\mathbf{r}_t, \mathbf{s}_t)$, the subspace basis $\mathbf{L}_t$ is updated by minimizing the surrogate function of empirical cost function $g_t(\mathbf{L}_t)$, which defines a optimization problem as

$$\begin{aligned}\mathbf{L}_t^* &= \arg\min_{\mathbf{L}_t} g_t(\mathbf{L}_t) \\ &= \arg\min_{\mathbf{L}_t} \operatorname{Tr}(\mathbf{L}_t^T(\lambda_1\mathbf{I} + \mathbf{A}_t)\mathbf{L}_t) - 2\operatorname{Tr}(\mathbf{L}_t^T\mathbf{B}_t)\end{aligned} \tag{16}$$

in which $\operatorname{Tr}(\cdot)$ denotes the trace of a matrix, and $\mathbf{A}_t$ and $\mathbf{B}_t$ are two auxiliary accumulation matrices that are introduced to remove duplicated calculations at each time instance

$$\begin{cases}\mathbf{A}_t = \mathbf{A}_{t-1} + \mathbf{r}_t\mathbf{r}_t^T \\ \mathbf{B}_t = \mathbf{A}_{t-1} + (\mathbf{d}_t - \mathbf{s}_t)\mathbf{r}_t^T.\end{cases} \tag{17}$$

Similar to [32], [33], and [41], the optimization problem defined in (16) is solved by a BCD Method for avoiding matrix inverse of large matrix. The subspace basis update algorithm is illustrated in Algorithm 3.

The subspace basis $\mathbf{L}_0$ is initialized before starting O-LSD. $\mathbf{L}_0$ can be initialized by either the first a few frames in the given sequence or their principal components [35], [45].

Algorithm 3 BCD Method for Subspace Basis Update

Input: $\mathbf{L}_{t-1} = [\mathbf{l}_1, \ldots, \mathbf{l}_r] \in \mathbb{R}^{p*r}$, $\mathbf{A}_t = [\mathbf{a}_1, \ldots, \mathbf{a}_r] \in \mathbb{R}^{r*r}$, $\mathbf{B}_t = [\mathbf{b}_1, \ldots, \mathbf{b}_r] \in \mathbb{R}^{p*r}$ and $\lambda_1 > 0$
Output: $\mathbf{L}_t$
1: $\tilde{\mathbf{A}} = \mathbf{A}_t + \lambda_1\mathbf{I}$.
2: **for** $i = 1$ to r **do**
3: $\mathbf{l}_i = \frac{1}{\tilde{\mathbf{A}}_{i,i}}(\mathbf{b}_i - \mathbf{L}_{t-1}\mathbf{a}_i) + \mathbf{l}_i$.
4: **end for**
5: $\mathbf{L}_t = [\mathbf{l}_1, \ldots, \mathbf{l}_r]$.
6: **return** $\mathbf{L}_t$

In satellite videos, moving objects move slowly and choosing these initialization scheme risks, including the slow-moving foreground objects into the background, which, thus, influences the detection performance, or a more extended sequence for initialization is required. Such a satellite video with adequate length is, however, not available technically yet. Therefore, in satellite videos, we recommend initializing $\mathbf{L}_0$ by random values instead, which performs pretty well in practice.

E. Convergence Analysis

One technical contribution of this article is to present the proposed O-LSD algorithm converges almost surely under mild condition.

Assumption 1: The observed data are uniformly bounded, and each data are independent.

As a widely used assumption [37], [40], [41], this assumption on the boundedness of observation data is quiet natural for real videos. Based the above-mentioned assumption, we present our first conclusion on the convergence of the surrogate function $g_t(\mathbf{L}_t)$.

Theorem II.2: *Let* $\{\mathbf{L}_t\}_{t=1}^{\infty}$ *be the sequence of solution obtained by Algorithm 1, the surrogate function* $g_t(\mathbf{L_t})$ *converges almost surely.*

Similarly, we obtain the convergence of two solutions obtained at two consecutive time instances by Algorithm 1.

Theorem II.3: *For two solutions produced by Algorithm 1 at two consecutive time instances,* $\|\mathbf{L}_t - \mathbf{L}_{t+1}\|_F = O(\frac{1}{t})$.

Then, we analyze the gap between the empirical cost function $f_t(\mathbf{L}_t)$ and its surrogate function $g_t(\mathbf{L}_t)$ with the estimated $\mathbf{L}_t$.

Theorem II.4: *Note that* $f_t(\mathbf{L})$ *is the empirical cost function defined in (6), and* $g_t(\mathbf{L})$ *is its surrogate function defined in (9).* $\mathbf{L}_t$ *is the solution obtained by Algorithm 1; when t tends to infinity,* $g_t(\mathbf{L}_t) - f_t(\mathbf{L}_t)$ *converges to* 0 *almost surely.*

In stochastic optimization, the expected cost function over $\mathbf{L}$ is defined as

$$f(\mathbf{L}) = \mathbb{E}_{\mathbf{d}}[\ell(\mathbf{d}, \mathbf{L})] = \lim_{t\to\infty} f_t(\mathbf{L}) \tag{18}$$

then we present the convergence of the gap between the expect cost function $f(\mathbf{L}_t)$ and the surrogate function $g_t(\mathbf{L}_t)$.

Theorem II.5: *As t tends to infinity, given the* $\mathbf{L}_t$ *is obtained by Algorithm 1,* $g_t(\mathbf{L}_t) - f(\mathbf{L}_t)$ *converges to* 0 *almost surely.*

Furthermore, the solution $\mathbf{L}_t$ obtained by Algorithm 1 is not a stationary point of expected cost function $f(\mathbf{L})$ when t tends to infinity, which, on contrary, is proved true

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TABLE II
INFORMATION ON THE EVALUATION DATA SETS

Video	Frame Size	Cross Validation		Performance Evaluation	
		#Frames	#Vehicles	#Frames	#Vehicles
001	400 × 400	200	9306	500	18167
002	600 × 400	200	13443	500	39362

in [32], [41], and [42]. Due to the existence of more than one solutions to (8), $\hat{\ell}(\mathbf{d}, \mathbf{L}, \mathbf{r}, \mathbf{s})$ is no longer strictly convex (or strongly convex) with respect to $(\mathbf{r}, \mathbf{s})$, and the gradient of the expected cost function $\nabla_{\mathbf{L}} f(\mathbf{L})$ is no longer Lipschitz. Therefore, the gradient of the expected cost function $\nabla_{\mathbf{L}} f(\mathbf{L})$ would not become zero when t tends to infinity, based on which we conclude the solution $\mathbf{L}_t$ may not be the stationary point of the expected cost function as t tends infinity.

Please refer to the Appendices for detailed proofs of the presented theorems.

F. Complexity Analysis

The complexity of OLD is determined by the iterative optimization in Algorithm 1. For each iteration, the complexity of foreground and background separations by Algorithm 2 is $\mathcal{O}(r^3 + \sum_{g\in\mathcal{G}} |g|)$, where $|g|$ is the number of elements in g, and the complexity for updating the subspace basis by (17) and Algorithm 3 is $\mathcal{O}(r^2 + pr + pr^2)$. This sums up to $\mathcal{O}(r^2 + pr + pr^2 + r^3 + \sum_{g\in\mathcal{G}} |g|)$ for each iteration. We observed that O-LSD convergences within a few iterations and $p \gg r$ in practice, so the overall complexity of O-LSD is $\mathcal{O}(pr^2 + \sum_{g\in\mathcal{G}} |g|)$.

For processing a video in the length of n, the complexity of O-LSD is $\mathcal{O}(n(pr^2 + \sum_{g\in\mathcal{G}} |g|))$, whereas the computation complexity of the batch-based LSD is $\mathcal{O}(n(p^2 + \sum_{g\in\mathcal{G}} |g|))$. In practice, a satellite video tends to cover a large territory by a considerable number of pixels; then, $p \gg r^2$ and $p^2 \gg pr^2$. Therefore, the complexity reduction by O-LSD is obvious, which makes O-LSD more suitable for large satellite video sequences.

III. EXPERIMENTS

The detection performance of O-LSD was evaluated on a data set of two satellite videos. This data set is constructed from a satellite video captured over Las Vagas, USA, on March 25, 2014, whose spatial resolution is 1.0 m and the frame rate is 30 frames/s. Both videos contain 700 frames with boundary boxes for moving vehicles as ground truth, and details on both videos are listed in Table II.[2] In this article, we used the first 200 frames in each video for the discussion on parameter selection, and the remaining frames were utilized for performance evaluation against existing state-of-the-art methods.

The detection performance on MOD is evaluated on recall, precision, and F_1 scores given by

$$\begin{aligned} \text{recall} &= \text{TP}/(\text{TP}+\text{FN}) \\ \text{precision} &= \text{TP}/(\text{TP}+\text{FP}) \\ F_1 &= \frac{2 \times \text{recall} \times \text{precision}}{\text{recall} + \text{precision}} \end{aligned} \tag{19}$$

[2]Moving vehicles are manually labeled by the Computer Vision Annotation Tool (CVAT), and a boundary box is provided for each moving object on each frame.

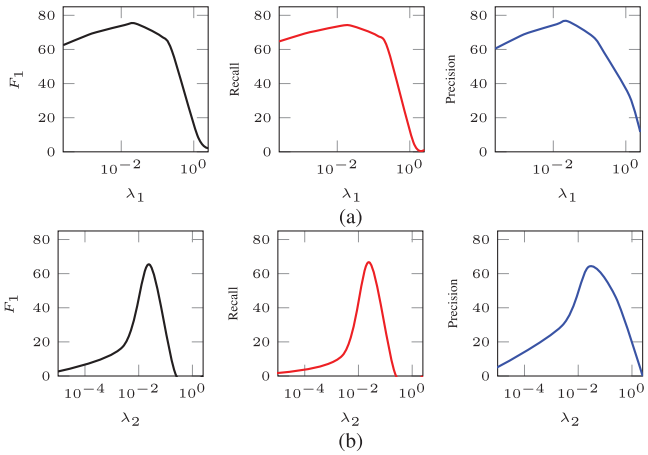


Fig. 1. MOD performance with different values of λ_1 and λ_2. (a) Varying λ_1 with $\lambda_2 = 0.025$. (b) Varying λ_2 with $\lambda_1 = 0.0025$.

where TP denotes the number of correct detections, and FN and FP are the numbers of missed detections and false alarms, respectively. In this article, we define a correct detection with maximum Intersection over Union (IoU) against the ground truth greater than a threshold. To complement the vehicles in small size in satellite videos, the threshold is set as 0.3.[3] In this article, we refer five-frame detection performance at each time instance to the metrics obtained from its five latest frames, which is used for observing the convergence, and the accumulated detection performance is measured on all frames before current time instances, which is used for comparing the overall performance over a sequence.

A. Parameter Setting

The performance of O-LSD is controlled by the dimension of the estimated subspace r and two weights for the low-rank term and the structured sparsity penalty separably, λ_1 and λ_2. The following experiments are conducted on the cross-validation sequence from Video 001.

The weight λ_1 assigns the importance of the low-rank subspace term. With the fixed λ_2, a significantly small λ_1 would encourage more information encoded in the low-rank subspace factors and hurt the detection performance in both terms of recall and precision. Increasing λ_1 improves the detection performance by the increased emphasis on the low-rank subspace modeling. After approaching the best detection performance, continuing increasing λ_1 would prevent the information encoded into the background. As illustrated in Fig. 1(a), with the fixed $\lambda_2 = 0.025$ and $r = 5$, as λ_1 increases, the F_1 score gradually increases to about 75% from around 60%, and then, the detection performance starts dropping with continuously increasing λ_1.

Increasing λ_2 with fixed λ_1 would put more emphasis on the structured sparsity of the extracted foreground, which, thus, improves the precision of the detected moving objects by restraining the random noises. When λ_2 continuously increases, the weight for the structured sparsity norm

[3]The estimated foreground is built by contiguous values rather than binary value, so we deploy threshold segmentation as postprocessing for extracting the foreground mask and the moving objects [49].

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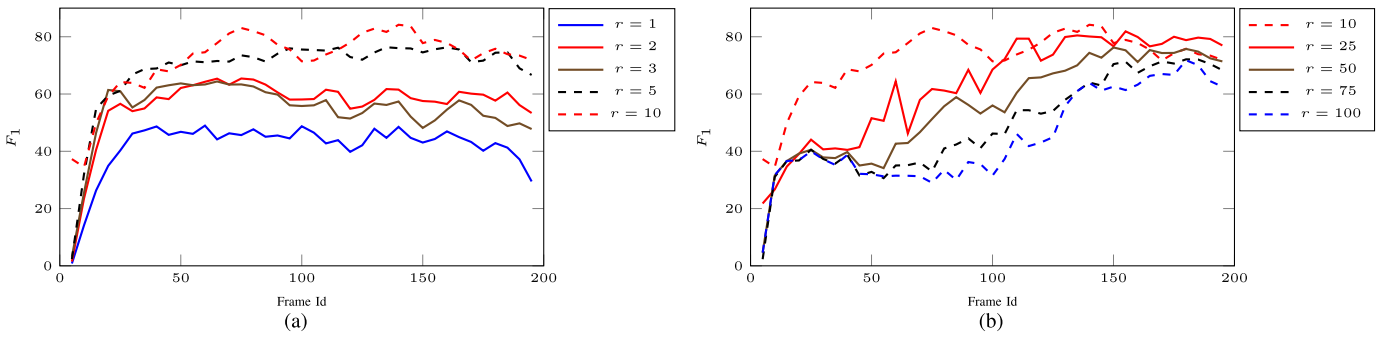


Fig. 2. Five-frame performance evaluation of O-LSD with a different value of r on cross-validation sequence from Video 001. (a) r from 1 to 10. (b) r from 10 to 100.

TABLE III
DETECTION PERFORMANCE COMPARISON WITH ONLINE ALGORITHMS

Video	001			002			Avg(F_1)
	Recall	Precision	F_1 score	Recall	Precision	F_1 score	
GRASTA	**76.96%**	31.76%	44.97%	72.22%	46.73%	56.75%	50.86%
OR-PCA	66.51%	41.50%	51.11%	71.94%	73.79%	72.86%	61.99%
GUSOS	59.35%	49.15%	53.77%	68.03%	62.61%	65.21%	59.49%
O-LSD	64.99%	**63.75%**	**64.36%**	**73.00%**	**90.21%**	**80.69%**	**72.48%**

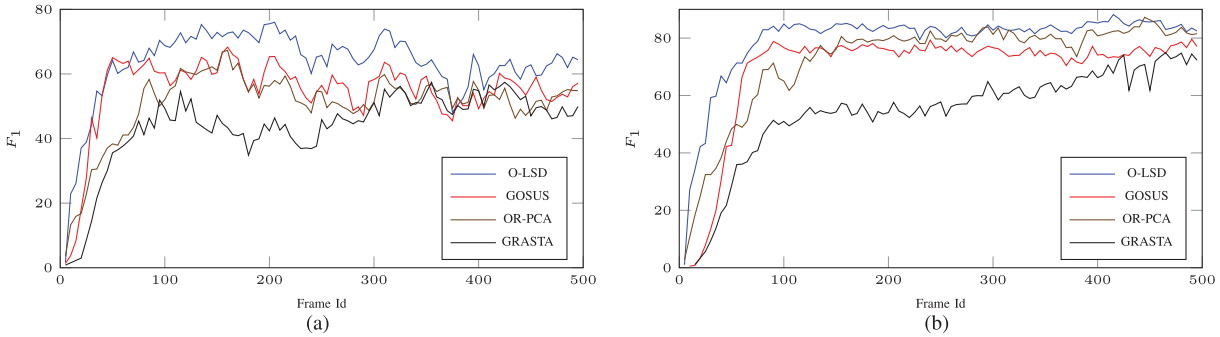


Fig. 3. Five-frame detection performance by different online algorithms in (a) Video 001 and (b) Video 002.

tends too large to encode information into the foreground, which then decreases the detection performance. As illustrated in Fig. 1(b), with fixed $\lambda_1 = 0.0025$ and $r = 5$, the F_1 score first approaches to the highest point as λ_2 increases, and then, the same metric drops when λ_2 tends too large.

Then, we discuss the selection of the dimension of the subspace r. With fixed λ_1 and λ_2, the O-LSD with smaller r probably converges faster; however, it may fail in modeling the permutation of the background for a long video sequence. On the contrary, selecting a higher r would disadvantage the updating of a subspace basis and require more frames before O-LSD converges. As shown in Fig. 2, for $r \leq 10$, the five-frame F_1 scores increase faster than those with $r \geq 25$. As the sequence length increases, five-frame F_1 scores by the O-LSD with $r \leq 10$ show a trend of dropping. For $r = 25$, this trend is negligible, and the same metric is still rising for $r > 25$, which means that the O-LSD requires a longer sequence to converge. The highest F_1 score at Frame-200 is achieved by $r = 25$, as shown in Fig. 2.

In the rest of this article, based on cross evaluation, we select $\lambda_1 = 1/\sqrt{p}$ and $\lambda_1/\lambda_2 = 0.1$ with $r = 25$ for evaluation, and further fine-tuning on the parameter selection would improve the detection performance by O-LSD.

B. Comparison With Online Approach

To verify the effectiveness of O-LSD, detection results by O-LSD are compared against the state-of-the-art online approaches, such as GRASTA [34], OR-PCA [37], and GOSUS [35]. For all methods, the subspace is initialized by the random scheme, and their parameters are selected through cross validation.

O-LSD boosts the detection performance in terms of the precision and F_1 scores. This improvement should be due to the structured sparsity penalty, which suppresses the random noises in estimated foreground frames. As presented in Table III, the O-LSD achieves the highest precision and F_1 scores on both videos although there is a little drop of recall score in Video 001 compared with GRASTA and OR-PCA. In terms of online methods employing structured sparsity penalty, O-LSD outperforms GOSUS on both satellite videos. For common MOD tasks, GOSUS is proved effective; however, it produces no improvement to OR-PCA on the satellite videos.

Another advantage of O-LSD is its faster convergence against other state-of-the-art online algorithms. As demonstrated in Fig. 3, in both videos, the O-LSD achieves the higher F_1 scores earlier than GRASTA, OR-PCA, and GOSUS,

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TABLE IV
DETECTION PERFORMANCE COMPARISON AGAINST BATCH-BASED ALGORITHMS

Video	001				002				Avg(F_1)
	Recall	Precision	F_1 score	Time Per Frame	Recall	Precision	F_1 score	Time Per Frame	
RPCA	**94.57%**	40.65%	56.86%	**3.16s**	**90.15%**	78.06%	83.67%	**5.45s**	70.27%
LSD	86.80%	**70.79%**	**77.98%**	68.48s	82.19%	**90.87%**	**86.31%**	119.50s	**82.15%**
O-LSD	64.99%	63.75%	64.36%	6.57s	73.00%	90.21%	80.69%	10.75s	72.48%

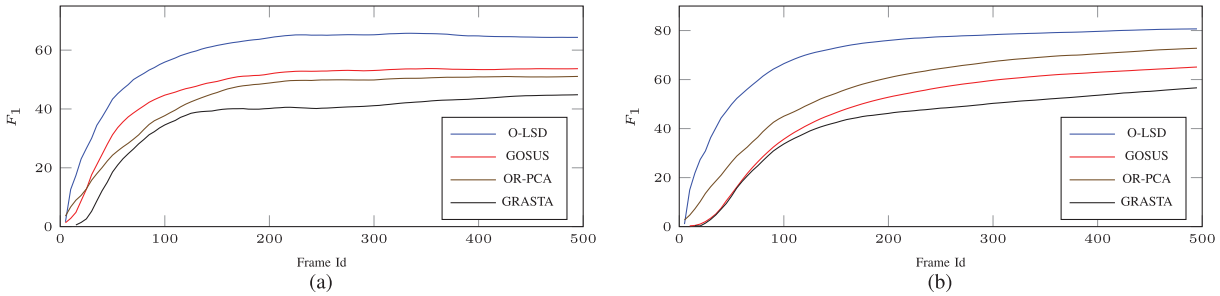


Fig. 4. Accumulated detection performance by different online algorithms in (a) Video 001 and (b) Video 002.

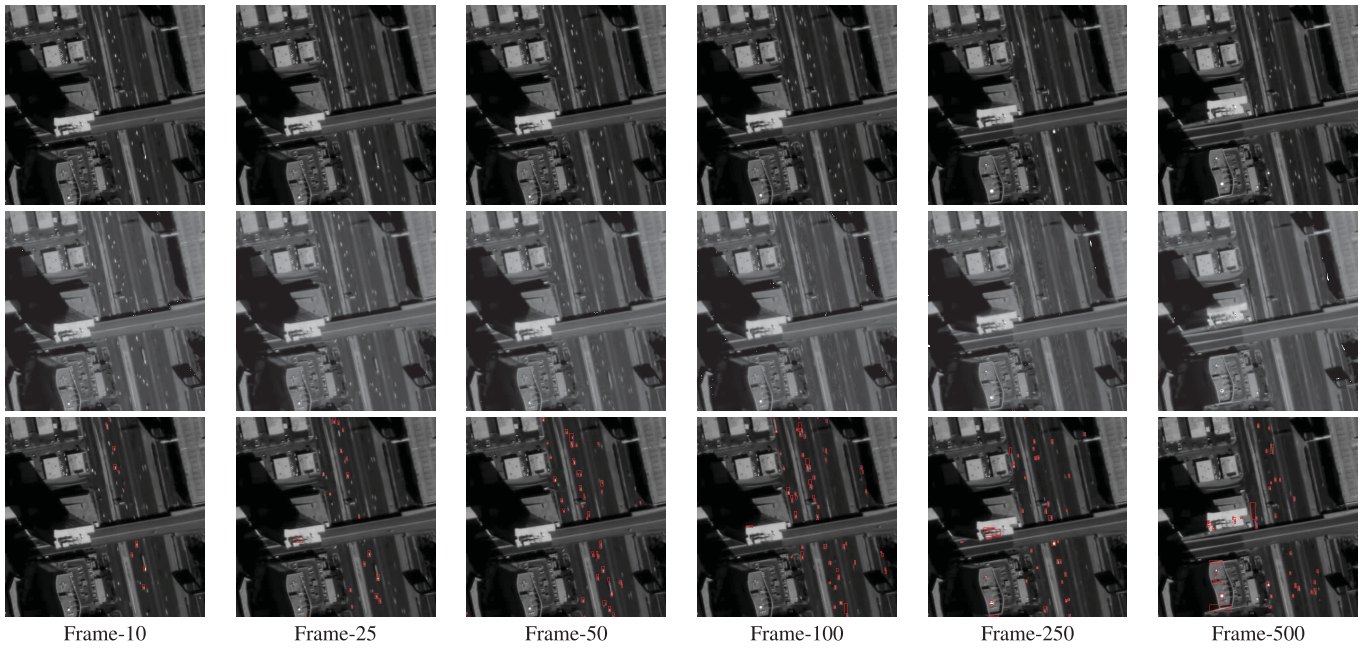


Fig. 5. Visualization of the estimated background and detection by O-LSD in Video 001. First row: original input. Second and third rows: estimated background frames and detection results, respectively.

implying that O-LSD converges faster by introducing the structured sparsity penalty. Similar trends can also be observed from the perspective of accumulated detection performance, as the accumulated detection performance of O-LSD is always better than the three existing methods as the length of sequence increases, as shown in Fig. 4.

Besides, as more frames are processed by the O-LSD algorithm, the cleanness of the estimated background frame gradually improves, as shown in Fig. 5.

C. Comparison With Batch-Based Approach

Besides the comparison against the state-of-the-art online algorithms, we also compare O-LSD with the batch-based algorithms that are RPCA [50] and LSD [29]. O-LSD achieves slightly descent detection performance with significantly reduced delay in processing on both videos.

Compared with batch-based approaches, O-LSD achieves comparable performance with the batch-based approach RPCA since it generates fewer false alarms, as shown in Fig. 6. O-LSD fails in matching the detection performance by its batch-based counterpart, LSD, and Table IV presents a drop of about 10% in F_1 score by O-LSD. This little gap in detection performance between LSD and O-LSD implies that O-LSD works pretty well although it may not converge to the global optimum of LSD.

In terms of the processing time for each frame, O-LSD significantly reduces this metric compared with LSD. As presented in Table IV, the time cost per frame for O-LSD is

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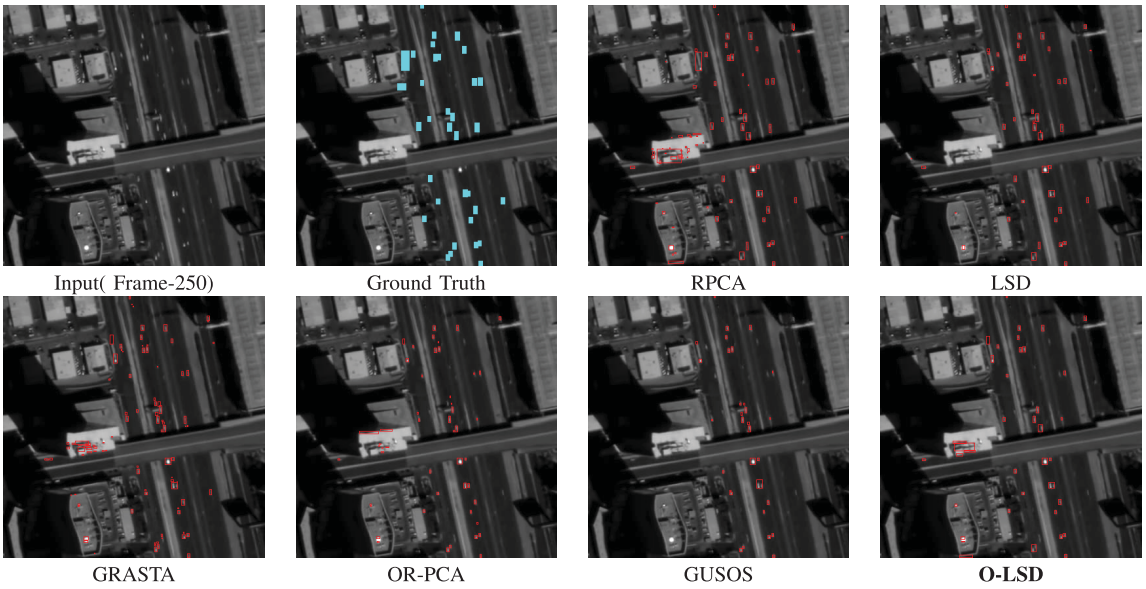


Fig. 6. Detection results obtained by different algorithms.

TABLE V
DETECTION PERFORMANCE BY O-LSD WITH TEMPORALLY DOWNSAMPLING

Video	T	Recall	Precision	F_1	Time per Frame
001	1	64.99%	63.75%	64.36%	6.57s
	3	67.85%	59.79%	63.57%	12.74s
	5	69.22%	62.02%	65.42%	14.62s
	10	67.72%	62.14%	64.91%	14.26s
002	1	73.00%	90.21%	80.69%	10.75s
	3	74.21%	88.46%	80.71%	14.48s
	5	74.41%	87.72%	80.52%	17.86s
	10	73.46%	88.68%	80.36%	21.12s

ten times smaller than LSD. Compared with RPCA, O-LSD improves the detection performance with moderately increased time cost per frame. As the detection results by those batch-based approaches are not available until the entire optimization is completed, O-LSD significantly reduces the delay in MOD by the O-LSD.

D. Performance Evaluation With Temporally Downsampling

In addition, we evaluate the effects of temporally downsampling on detection performance. In this set of experiments, one frame from every T frames, $T \in \{1, 3, 5, 10\}$, is fed to O-LSD.

As shown in Table V, with increasing T from 1 to 10, the F_1 scores fluctuate negligible, which implies that detection performance of O-LSD is almost not influenced by the temporary downsampling. At the same time, the time costs for each frame are more or less the same with different temporally downsampling frequencies. Since fewer frames need to be processed with large temporally downsampling scales, the delay for processing is reduced. However, considering that it takes more than 1 s to obtain detection results for each frame by O-LSD, it is still hard to directly apply O-LSD for real-time applications without appropriate accelerations.

IV. CONCLUSION

The main contribution of this article is an effective algorithm, O-LSD, which combines the stochastic optimization and structured sparsity penalty to improve online subspace estimation method for MOD in satellite videos. We elaborate on the model of O-LSD and its optimization method that is proved to converge almost surely under mild condition. The experiments on a data set of two satellite videos validate the improvement of O-LSD to the existing state-of-the-art approaches. With the temporal downsampling scheme, O-LSD also reduces the processing delay with almost unchanged performance.

APPENDIX A
TECHNICAL LEMMA

Lemma A.1 (Danskin's Theorem From [51]): *Let $\mathcal{C} \subset \mathbb{R}^m$ be a compact set. The function $\ell(\mathbf{x}, \mathbf{u}) : \mathbb{R}^n \times \mathcal{C} \to \mathcal{R}$ is continuous, and $\ell(\cdot, \mathbf{u})$ is convex with regards to $\mathbf{x}$ for every $u \in \mathcal{C}$. Define $\ell(\mathbf{x}) = \min_{\mathbf{u}\in\mathcal{C}} \ell(\mathbf{x}, \mathbf{u})$ and $\mathcal{C}(\mathbf{x}) = \{\mathbf{u}^* | \mathbf{u}^* = \arg\min_{\mathbf{u}} \ell(\mathbf{x}, \mathbf{u})\}$.*

If $\ell(\mathbf{x}, \mathbf{u})$ is differentiable with respect to $\mathbf{x}$ for all $\mathbf{u} \in \mathcal{C}$, and $(\partial\ell(\mathbf{x}, \mathbf{u})/\partial\mathbf{x})$ is continuous with respect to $\mathbf{u}$ for all $\mathbf{x}$, then the subgradient of $\ell(\mathbf{x})$ is given by

$$\partial_{\mathbf{x}}\ell(\mathbf{x}) = \text{conv}\left\{\frac{\partial\ell(\mathbf{x}, \mathbf{u})}{\partial\mathbf{x}} | \mathbf{u} \in \mathcal{C}(\mathbf{x})\right\} \tag{20}$$

where conv$\{\cdot\}$ *indicates the convex hull operator.*

Lemma A.2 ([52], Lemma 2.6): *Let $f : \mathbf{X} \to \mathbb{R}$ be a convex function. Then, f is L-Lipschitz over $\mathbf{X}$ with respect to a norm $\|\cdot\|$ if and only if for all $\forall\mathbf{w} \in \mathbf{X}$ and $\mathbf{z} \in \partial f(\mathbf{w})$, we have that $\|\mathbf{z}\|_* \leq L$, where $\|\cdot\|_*$ is the dual norm.*

Lemma A.3 (Sufficient Condition of Convergence for a Stochastic Optimization From [52]): *Let $(\Omega, \mathcal{F}, P)$ be a measurable probability space, μ_t, for $t > 0$, be the realization of a stochastic process, and $\mathcal{F}_t$ be the filtration by the past*

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information at time t. Let

$$\delta_t = \begin{cases} 1, & \text{if } \mathbb{E}[u_{t+1} - u_t | \mathcal{F}_t] > 0 \\ 0, & \text{otherwise.} \end{cases} \tag{21}$$

If for all t, $\mu_t \geq 0$ and $\sum_{t=1}^{\infty} \mathbb{E}[\delta_t(u_{t+1} - u_t)] < \infty$, then μ_t is a quasi-martingale and converges almost surely. Moreover

$$\sum_{t=1}^{\infty} |\mathbb{E}[u_{t+1} - u_t | \mathcal{F}_t]| < +\infty \text{ a.s. .} \tag{22}$$

Lemma A.4 (Corollary of Donsker Theorem From [53]): *Let $\mathbf{F} = \{f_\theta : \mathcal{X} \to \mathbb{R}, \theta \in \Theta\}$ be a set of measurable functions indexed by a bounded subset Θ of $\mathbb{R}^d$. Suppose that there exists a constant K such that*

$$\left| f_{\theta_1} - f_{\theta_2} \right| \leq K \|\theta_1 - \theta_2\|_2 \tag{23}$$

for every θ_1 and θ_2 in Θ and $\mathbf{x}$ in $\mathcal{X}$. Then, F is P-Donsker. For any f in F, let us define $\mathbb{P}_n f$, $\mathbb{P} f$, and $\mathbb{G}_n f$ as

$$\begin{aligned} \mathbb{P}_n f &= \frac{1}{n} \sum_{i=1}^{n} f(\mathbf{X}_i) \\ \mathbb{P} f &= \mathbb{E}[f(X)] \\ \mathbb{G}_n f &= \sqrt{n}(\mathbb{P}_n f - \mathbb{P} f). \end{aligned} \tag{24}$$

Let us also suppose that for all $\mathbb{P}$, $f^2 \leq \delta^2$ and $\|f\|_\infty \leq M$ and that the random variables $\mathbf{X}_1, \mathbf{X}_2, \ldots$ are Borel-measurable. Then, we have

$$\mathbb{E}|\mathbb{G}_n f|_F = O(1) \tag{25}$$

where $|\mathbb{G}_n f|_F = \sup_{f \in F} |\mathbb{G}_n f|$.

Lemma A.5 (Positive Converging Sums From [41]): *Let a_n and b_n be two real sequences such that for all $n, a_n \geq 0$, $b_n \geq 0$, $\sum_{n=1}^{\infty} a_n = \infty$, and $\sum_{n=1}^{\infty} a_n b_n < \infty$, $\exists K > 0$ s.t. $|b_{n+1} - b_n| < K a_n$. Then, $\lim_{n \to \infty} b_n = 0$.*

APPENDIX B
PROPOSITION

Proposition B.1: *Assume that $\mathbf{d} \in \mathbf{D}$ is uniformly bounded, and $(\mathbf{r}^*, \mathbf{s}^*)$ is the minimizers of the reconstruction cost function $\hat{\ell}(\mathbf{d}, \mathbf{L}, \mathbf{r}, \mathbf{s})$ obtained by Algorithm 2. Then, the following holds.*

1) *$\mathbf{r}^*$ and $\mathbf{s}^*$ are uniformly bounded.*
2) *$(1/t)\mathbf{A}_t$ and $(1/t)\mathbf{B}_t$ are uniformly bounded.*
3) *$\mathbf{L}_t$ is supported by a compact subset $\mathcal{L}$.*

Proof: Given $(\mathbf{0}, \mathbf{d})$ is a nontrivial feasible solution to (7), for the optimal solution $(\mathbf{r}^*, \mathbf{s}^*)$

$$\begin{aligned} \frac{1}{2}\left\|\mathbf{d} - \mathbf{L}\mathbf{r}^* - \mathbf{s}^*\right\|_2^2 + \frac{\lambda_1}{2}\left\|\mathbf{r}^*\right\|_2^2 + \lambda_2 \left\|\mathbf{s}^*\right\|_{\ell_1/\ell_\infty} \\ \leq \hat{\ell}(\mathbf{d}, \mathbf{L}, \mathbf{0}, \mathbf{d}) \leq \lambda_2 \|\mathbf{d}\|_{\ell_1/\ell_\infty} \end{aligned} \tag{26}$$

and thus, we obtain that

$$\begin{aligned} \left\|\mathbf{r}^*\right\|_2^2 &\leq \frac{2\lambda_2}{\lambda_1} \|\mathbf{d}\|_{\ell_1/\ell_\infty} \\ \left\|\mathbf{s}^*\right\|_{\ell_1/\ell_\infty} &\leq \|\mathbf{d}\|_{\ell_1/\ell_\infty}. \end{aligned} \tag{27}$$

Based on the assumption that $\mathbf{d}$ is uniformly bounded, $\mathbf{r}^*$ and $\mathbf{s}^*$ are uniformly bounded.

Similarly, we show that the accumulation matrices $\mathbf{A}_t$ and $\mathbf{B}_t$ are also uniformly bounded, as

$$\begin{aligned} \frac{1}{t}\mathbf{A}_t &= \frac{1}{t}\sum_{i=1}^{t} \mathbf{r}_i \mathbf{r}_i^T \\ \frac{1}{t}\mathbf{B}_t &= \frac{1}{t}\sum_{i=1}^{t} (\mathbf{d}_i - \mathbf{s}_i)\mathbf{r}_i{}^T. \end{aligned} \tag{28}$$

The closed-from solution $\mathbf{L}_t$ is given as

$$\begin{aligned} \mathbf{L}_t &= \mathbf{B}_t(\mathbf{A}_t + \lambda_1 \mathbf{I})^{-1} \\ &= \frac{1}{t}\mathbf{B}_t \left(\frac{1}{t}\mathbf{A}_t + \frac{\lambda_1}{t}\mathbf{I}\right)^{-1} \end{aligned} \tag{29}$$

in which $(1/t)\mathbf{A}_t$ and $(1/t)\mathbf{B}_t$ are uniformly bounded; therefore, $\mathbf{L}_t$ is uniformly bounded. ∎

Proposition B.2: *Let $\mathbf{r}$, $\mathbf{s}$, and $\mathbf{L}_t$ be the solution obtained by Algorithm 1. The following properties can be obtained:*

1) *$\hat{\ell}(\mathbf{d}, \mathbf{L}, \mathbf{r}, \mathbf{s})$ and $\ell(\mathbf{d}, \mathbf{L})$ are uniformly bounded.*
2) *The surrogate function $g_t(\mathbf{L})$ is uniformly bounded and Lipschitz.*

Proof: The first claim is proved by combining the definition of $\hat{\ell}(\mathbf{d}, \mathbf{L}, \mathbf{r}, \mathbf{s})$ and the uniform boundedness of $\mathbf{d}$, $\mathbf{L}$, $\mathbf{r}$, and $\mathbf{s}$. Similarly, we can show that $g_t(\mathbf{L})$ is uniformly bounded.

To prove that $g_t(\mathbf{L})$ is Lipschitz, we show that the gradient of $g_t(\mathbf{L})$ is uniformly bounded as

$$\begin{aligned} \|\nabla g_t(\mathbf{L})\|_F &= \left\| \mathbf{L}\left(\frac{1}{t}\mathbf{A}_t - \frac{1}{t}\mathbf{I}\right) - \frac{1}{t}\mathbf{B}_t \right\|_F \\ &\leq \|\mathbf{L}\|_F \left(\left\| \frac{1}{t}\mathbf{A}_t \right\|_F + \left\| \frac{1}{t}\mathbf{I} \right\|_F \right) + \left\| \frac{1}{t}\mathbf{B}_t \right\|_F \end{aligned} \tag{30}$$

where the terms on the right-hand side of the inequality are uniformly bounded, and $\|\nabla g_t(\mathbf{L})\|_F$ is uniformly bounded. According to Lemma A.2, $g_t(\mathbf{L})$ is convex with respect to $\mathbf{L}$, and the boundedness of the gradient implies that $g_t(\mathbf{L})$ is Lipschitz. ∎

Proposition B.3: *$\mathcal{X}_0(\mathbf{L})$ refers to the set of all minimizers to $\hat{\ell}(\mathbf{d}, \mathbf{L}, \mathbf{r}, \mathbf{s})$ as*

$$\mathcal{X}_0(\mathbf{L}) = \{(\overline{\mathbf{r}}, \overline{\mathbf{s}}) | (\overline{\mathbf{r}}, \overline{\mathbf{s}}) = \arg\min_{\mathbf{r}, \mathbf{s}} \hat{\ell}(\mathbf{d}, \mathbf{L}, \mathbf{r}, \mathbf{s})\}. \tag{31}$$

1) *The subgradient of function $\ell(\mathbf{d}, \mathbf{L})$ with respect to $\mathbf{L}$ is given as*

$$\begin{aligned} \partial_L \ell(\mathbf{d}, \mathbf{L}) = \text{conv}\{(\mathbf{L}\mathbf{r}^* + \mathbf{s}^* - \mathbf{d})\mathbf{r}^{*T} \\ | (\mathbf{r}^*, \mathbf{s}^*) \in \mathcal{X}_0(\mathbf{L})\} \end{aligned} \tag{32}$$

   where conv$\{\cdot\}$ *is a convex hull operator.*
2) *The subgradient $\partial_L \ell(\mathbf{d}, \mathbf{L})$ is uniformly bounded, and $\ell(\mathbf{d}, \mathbf{L})$ is uniformly Lipschitz.*

Proof: To the best of our knowledge, there is no available necessary and sufficient condition for uniqueness of minimizer to the reconstruction cost function, which implies that more than one minimizer of $\hat{\ell}(\mathbf{d}, \mathbf{L}, \mathbf{r}, \mathbf{s})$ probably exist.

$\hat{\ell}(\mathbf{z}, \mathbf{L}, \mathbf{r}, \mathbf{s})$ is convex and differentiable with respect to $\mathbf{L}$ for every feasible $(\mathbf{r}^*, \mathbf{s}^*)$. Given $(\partial \hat{\ell}(\mathbf{z}, \mathbf{L}, \mathbf{r}, \mathbf{s})/\partial \mathbf{L}) = (\mathbf{L}\mathbf{r} + \mathbf{s} - \mathbf{d})\mathbf{r}^T$ is differentiable with respect to $(\mathbf{r}, \mathbf{s})$ for all $\mathbf{L}$, according to Lemma A.1, the subgradient of $\ell(\mathbf{d}, \mathbf{L})$ is given as

$$\partial_L \ell(\mathbf{d}, \mathbf{L}) \in \text{conv}\{(\mathbf{L}\mathbf{r}^* + \mathbf{s}^* - \mathbf{d})\mathbf{r}^{*T} | (\mathbf{r}^*, \mathbf{s}^*) \in \mathcal{Z}_0(\mathbf{L})\}. \tag{33}$$

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Then, we proof that $(\mathbf{Lr}+\mathbf{s}-\mathbf{d})\mathbf{r}^T$ is uniformly bounded

$$\left\|(\mathbf{Lr}+\mathbf{s}-\mathbf{d})\mathbf{r}^T\right\|_2 \leq \|\mathbf{r}\|_2(\|\mathbf{L}\|_F\|\mathbf{r}\|_2+\|\mathbf{s}\|_2+\|\mathbf{d}\|_2) \quad (34)$$

in which every item on the right-hand side of the inequality is uniformly bounded. Therefore, $(\mathbf{Lr}+\mathbf{s}-\mathbf{d})\mathbf{r}^T$ is uniformly bounded. The convex hull of the bounded set is also bounded; thus, the subgradient $\partial_L \ell(\mathbf{d},\mathbf{L})$ is bounded.

By Lemma A.2, $\ell(\mathbf{d},\mathbf{L})$ is convex with respect to $\mathbf{L}$, and its subgradient is is uniformly bounded; thus, $\ell(\mathbf{d},\mathbf{L})$ is uniformly Lipschitz. ∎

Proposition B.4: *The empirical cost function $f_t(\mathbf{L})$ is uniformly bounded and Lipschitz.*

Proof: By checking the definition of the empirical cost function $f_n(\mathbf{L})$, $f_n(\mathbf{L})$ is also uniformly bounded.

For $\mathbf{g}_i \in \partial_L \ell(\mathbf{d}_i,\mathbf{L})$, the subgradient $\|\partial_{\mathbf{L}}\|_F$ is uniformly bounded

$$\|\partial_{\mathbf{L}}\|_F = \left\|\frac{1}{n}\sum_{i=1}^{n}\mathbf{g}_i+\frac{\lambda_1}{2n}\mathbf{L}\right\|_F \leq \frac{1}{n}\sum_{i=1}^{n}\|\mathbf{g}_i\|_F+\frac{\lambda_1}{2n}\|\mathbf{L}\|_F. \quad (35)$$

Because $f_n(\mathbf{L})$ is convex with respect to $\mathbf{L}$, and its subgradient $\|\partial_{\mathbf{L}}\|_F$ is uniformly bounded, $f_n(\mathbf{L})$ is Lipschitz. ∎

APPENDIX C
PROOF DETAILS

Theorem C.1: *Let $\{\mathbf{L}_t\}_{t=1}^{\infty}$ be the sequence of solution obtained by Algorithm 1, and the surrogate function $g_t(\mathbf{L_t})$ converges almost surely.*

Proof: In stochastic optimization, the expected cost function is defined all the samples

$$f(\mathbf{L}) = \mathbb{E}_{\mathbf{d}}[\ell(\mathbf{d},\mathbf{L})] = \lim_{n\to\infty} f_n(\mathbf{L}). \quad (36)$$

$g_t(\mathbf{L}_t)$ is analyzed as a stochastic positive process since each term in it is nonnegative and samples are drawn randomly (independent).

Note that $u_t = g_t(\mathbf{L}_t)$, and the difference between two consecutive time instances is given as

$$\begin{aligned} u_{t+1}-u_t &= g_{t+1}(\mathbf{L}_{t+1})-g_t(\mathbf{L}_t) \\ &= g_{t+1}(\mathbf{L}_{t+1})-g_{t+1}(\mathbf{L}_t)+g_{t+1}(\mathbf{L}_t)-g_t(\mathbf{L}_t) \\ &= g_{t+1}(\mathbf{L}_{t+1})-g_{t+1}(\mathbf{L}_t) \\ &\quad +\frac{\ell(\mathbf{d}_{t+1},\mathbf{L}_t)-f_t(\mathbf{L}_t)}{t+1}+\frac{f_t(\mathbf{L}_t)-g_t(\mathbf{L}_t)}{t+1}. \end{aligned} \quad (37)$$

The first two terms satisfy $g_{t+1}(\mathbf{L}_{t+1}) \leq g_{t+1}(\mathbf{L}_t)$ and $f_t(\mathbf{L}_t)-g_t(\mathbf{L}_t) \leq 0$; therefore

$$\begin{aligned} u_{t+1}-u_t &\leq \frac{\ell(\mathbf{d}_{t+1},\mathbf{L}_t)-f_t(\mathbf{L}_t)}{t+1} \\ &\leq \frac{\ell(\mathbf{d}_{t+1},\mathbf{L}_t)-\frac{1}{t}\sum_{i=1}^{t}\ell(\mathbf{d}_t,\mathbf{L}_t)}{t+1}. \end{aligned} \quad (38)$$

The expectation conditioned on past information $\mathcal{F}_t$ is given as

$$\begin{aligned} \mathbb{E}[u_{t+1}-u_t|\mathcal{F}_t] &\leq \frac{\mathbb{E}[\ell(\mathbf{d}_{t+1},\mathbf{L}_t)|\mathcal{F}_t]-\frac{1}{t}\sum_{i=1}^{t}\ell(\mathbf{d}_t,\mathbf{L}_t)}{t+1} \\ &\leq \frac{f(\mathbf{L}_t)-\frac{1}{t}\sum_{i=1}^{t}\ell(\mathbf{d}_t,\mathbf{L}_t)}{t+1} \\ &\leq \frac{\|f-f_t\|_\infty}{t+1} \end{aligned} \quad (39)$$

where we define $f_t = (1/t)\sum_{i=1}^{t}\ell(\mathbf{d}_t,\mathbf{L}_t)$, $f = \mathbb{E}_{\mathbf{d}}[\ell(\mathbf{d},\mathbf{L}_t)]$, and $\|f-f_t\|_\infty = \sup_{f\in\mathcal{F}}|f-f_t|$. $F = \{\ell(\mathbf{d},\mathbf{L}) : \mathcal{D} \to \mathbb{R}, \mathbf{L} \in \mathcal{L}\}$ defines a set of measurement function indexed by $\mathbf{L}_t$ from a compact subset $\mathcal{L}$ and P-Donsker. In addition, the boundedness of $\ell(\mathbf{d},\mathbf{L})$ implies that $\mathbb{E}[\ell(\mathbf{d},\mathbf{L})^2]$ is uniformly bounded. Then, the requirements of Lemma A.4 are all satisfied such that

$$\mathbb{E}[\sqrt{t}\|f-f_t\|_\infty] \leq \kappa, \kappa > 0. \quad (40)$$

Therefore

$$\begin{aligned} \mathbb{E}[[\mathbb{E}[u_{t+1}-u_t|\mathcal{F}_t]]^+] &= \mathbb{E}[\max\{0,\mathbb{E}[u_{t+1}-u_t|\mathcal{F}_t]\}] \\ &\leq \frac{\kappa}{t^{\frac{3}{2}}} \end{aligned} \quad (41)$$

where $[\cdot]^+$ is the positive variation operator.

Then, we deploy Lemma A.3 to present the convergence of $g_t(\mathbf{x})$. We define that

$$\delta_t = \begin{cases} 1, & \text{if } \mathbb{E}[u_{t+1}-u_t|\mathcal{F}_t] > 0 \\ 0, & \text{otherwise.} \end{cases} \quad (42)$$

We have

$$\begin{aligned} \sum_{t=1}^{\infty}\mathbb{E}[\delta_t(u_{t+1}-u_t)|\mathcal{F}_t] &= \sum_{t=1}^{\infty}\mathbb{E}[[\mathbb{E}[u_{t+1}-u_t|\mathcal{F}_t]]^+] \\ &\leq \sum_{t=1}^{\infty}\frac{\kappa}{t^{\frac{3}{2}}} < +\infty. \end{aligned} \quad (43)$$

Conclusively, $g_t(\mathbf{L}_t)$ is quasi-martingale and converges almost sure. Moreover

$$\sum_{t=1}^{\infty}|\mathbb{E}[u_{t+1}-u_t|\mathcal{F}_t]| < +\infty \text{ a.s..} \quad (44)$$

∎

Theorem C.2: *For two solutions produced by Algorithm 1 at two consecutive time instances*

$$\|\mathbf{L}_t-\mathbf{L}_{t+1}\|_F = O\left(\frac{1}{t}\right). \quad (45)$$

Proof: The Hessian matrix of $g_t(\mathbf{L})$ is $\mathbf{H} = \mathbf{I}\otimes(\mathbf{A_t}+\lambda_1\mathbf{I})$, where $\otimes$ is the Kronecker production operator. By the definition of $\mathbf{A}_t$, $\mathbf{A_t}$ is always semipositive; therefore, the smallest eigenvalue of $\mathbf{H}$ is greater than λ_1, which implies that $g_t(\mathbf{L})$ is strictly convex (perhaps strongly convex) with respect to $\mathbf{L}$ and

$$g_t(\mathbf{L}_{t+1})-g_t(\mathbf{L}_t) \geq \lambda_1\|\mathbf{L}_{t+1}-\mathbf{L}_t\|_F^2. \quad (46)$$

Because $\mathbf{L}_{t+1}$ minimizes $g_{t+1}(\mathbf{L})$, $g_{t+1}(\mathbf{L}_{t+1})-g_{t+1}(\mathbf{L}_t) < 0$, which presents that

$$\begin{aligned} g_t(\mathbf{L}_{t+1})-g_t(\mathbf{L}_t) &= g_t(\mathbf{L}_{t+1})-g_{t+1}(\mathbf{L}_{t+1})+g_{t+1}(\mathbf{L}_{t+1})-g_t(\mathbf{L}_t) \\ &\leq g_t(\mathbf{L}_{t+1})-g_{t+1}(\mathbf{L}_{t+1})+g_{t+1}(\mathbf{L}_t)-g_t(\mathbf{L}_t) \\ &\leq (g_t(\mathbf{L}_{t+1})-g_{t+1}(\mathbf{L}_{t+1}))-(g_t(\mathbf{L}_t)-g_{t+1}(\mathbf{L}_t)). \end{aligned} \quad (47)$$

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We define $G_t(\mathbf{L}) = g_t(\mathbf{L}) - g_{t+1}(\mathbf{L})$, and then, the gradient of $G_t(\mathbf{L})$ is extracted as

$$\begin{aligned}\nabla G_t(\mathbf{L}) &= \nabla g_t(\mathbf{L}) - \nabla g_{t+1}(\mathbf{L}) \\ &= \frac{1}{t}\mathbf{L}\hat{\mathbf{A}}_t + \frac{1}{t}\mathbf{B}_t - \frac{1}{t+1}\mathbf{L}\hat{\mathbf{A}}_{t+1} - \frac{1}{t+1}\mathbf{B}_{t+1} \\ &= \frac{1}{t}\mathbf{L}\left(\hat{\mathbf{A}}_t - \frac{t}{t+1}\hat{\mathbf{A}}_{t+1}\right) + \frac{1}{t}\left(\hat{\mathbf{B}}_t - \frac{t}{t+1}\hat{\mathbf{A}}_{t+1}\right)\end{aligned} \tag{48}$$

in which $\hat{\mathbf{A}}_t = \mathbf{A}_t + \lambda_1\mathbf{I}$.

Given that $\mathbf{A}_t$, $\mathbf{B}_t$, and $\mathbf{L}$ are uniformly bounded, the gradient $\nabla G_t(\mathbf{L})$ is also uniformly bounded

$$\begin{aligned}&\|\nabla G_t(\mathbf{L})\|_F \\ &\quad\le \frac{1}{t}\left(\|\mathbf{L}\|_F\left\|\hat{\mathbf{A}}_t - \frac{t}{t+1}\hat{\mathbf{A}}_{t+1}\right\|_F + \left\|\hat{\mathbf{B}}_t - \frac{t}{t+1}\hat{\mathbf{A}}_{t+1}\right\|_F\right).\end{aligned} \tag{49}$$

Thus, we conclude that $G_t(\mathbf{L})$ is uniformly Lipschitz with respect to $\mathbf{L}$, and there exists a positive constant $\kappa_t = 1/t(\|\mathbf{L}\|_F\|\hat{\mathbf{A}}_t - t/t+1\ \hat{\mathbf{A}}_{t+1}\|_F + \|\hat{\mathbf{B}}_t - t/t+1\ \hat{\mathbf{A}}_{t+1}\|_F)$ that satisfies

$$G_t(\mathbf{L}_t) - G_t(\mathbf{L}_{t+1}) \le \kappa_t\|\mathbf{L}_{t+1} - \mathbf{L}_t\|_F. \tag{50}$$

Combining (46), (47), and (50), we conclude that

$$\|\mathbf{L}_{t+1} - \mathbf{L}_t\|_F \le \frac{\kappa_t}{\lambda_1} \tag{51}$$

and $\|\mathbf{L}_{t+1} - \mathbf{L}_t\|_F = O(1/t)$. ■

Theorem C.3: *Note that $f_t(\mathbf{L})$ is the empirical cost function, and $g_t(\mathbf{L})$ is its surrogate function. $\mathbf{L}_t$ is the solution obtained by Algorithm 1, when t tends to infinity, $g_t(\mathbf{L}_t) - f_t(\mathbf{L}_t)$ converges to 0 almost surely.*

Proof: This proof is originally presented in [37]; for the completeness of this proof, we introduce it here. Combining (37) with $g_{t+1}(\mathbf{L}_{t+1}) - g_{t+1}(\mathbf{L}_t) \le 0$, we obtain

$$\begin{aligned}&\frac{g_t(\mathbf{L}_t) - f_t(\mathbf{L}_t)}{t+1} \\ &\quad\le \frac{\ell(\mathbf{d}_{t+1}, \mathbf{L}_t) - f_t(\mathbf{L}_t)}{t+1} - (g_{t+1}(\mathbf{L}_{t+1}) - g_t(\mathbf{L}_t)) \\ &\quad\le \frac{\ell(\mathbf{d}_{t+1}, \mathbf{L}_t) - f_t(\mathbf{L}_t)}{t+1} + [g_{t+1}(\mathbf{L}_{t+1}) - g_t(\mathbf{L}_t)]^-\end{aligned} \tag{52}$$

in which $[\cdot]^-$ refers to the negative variation operator.

Similar to the proof of Theorem C.1, we take the expectation conditioned on past information $\mathcal{F}_t$

$$\begin{aligned}\mathbb{E}\left[\frac{g_t(\mathbf{L}_t) - f_t(\mathbf{L}_t)}{t+1}\middle|\mathcal{F}_t\right] &= \frac{g_t(\mathbf{L}_t) - f_t(\mathbf{L}_t)}{t+1} \\ &\le \mathbb{E}\left[\frac{\ell(\mathbf{d}_{t+1}, \mathbf{L}_t) - f_t(\mathbf{L}_t)}{t+1}\middle|\mathcal{F}_t\right] \\ &\quad + \mathbb{E}[[g_{t+1}(\mathbf{L}_{t+1}) - g_t(\mathbf{L}_t)]^-|\mathcal{F}_t].\end{aligned} \tag{53}$$

Accumulating $(g_t(\mathbf{L}_t) - f_t(\mathbf{L}_t))/(t+1)$ with t tending to ∞, we have

$$\begin{aligned}&\sum_{t=1}^{\infty}\frac{g_t(\mathbf{L}_t) - f_t(\mathbf{L}_t)}{t+1} \\ &\quad\le \sum_{t=1}^{\infty}\mathbb{E}\left[\frac{\ell(\mathbf{d}_{t+1}, \mathbf{L}_t) - \frac{1}{t}\sum_{i=1}^{t}\ell(\mathbf{d}_t, \mathbf{L}_t)}{t+1}|\mathcal{F}_t\right] \\ &\qquad + \sum_{t=1}^{\infty}\mathbb{E}[[g_{t+1}(\mathbf{L}_{t+1}) - g_t(\mathbf{L}_t)]^-|\mathcal{F}_t] \\ &\quad\le \sum_{t=1}^{\infty}\frac{|f - f_t|}{t+1} + \sum_{t=1}^{\infty}\mathbb{E}[[g_{t+1}(\mathbf{L}_{t+1}) - g_t(\mathbf{L}_t)]^-|\mathcal{F}_t].\end{aligned} \tag{54}$$

According to central limit theorem, $\sqrt{t}|f - f_t|$ converges almost surely as t tends to infinity. From Theorem C.1, we also obtain that

$$\sum_{t=1}^{\infty}|\mathbb{E}[[g_{t+1}(\mathbf{L}_{t+1}) - g_t(\mathbf{L}_t)]^-|\mathcal{F}_t]| < +\infty. \tag{55}$$

Hence, we have the almost sure convergence of the positive sum

$$\begin{aligned}&\sum_{t=1}^{\infty}\frac{g_t(\mathbf{L}_t) - f_t(\mathbf{L}_t)}{t+1} \\ &\quad\le \sum_{t=1}^{\infty}\frac{|f - f_t|}{t+1} + \sum_{t=1}^{\infty}\mathbb{E}[[g_{t+1}(\mathbf{L}_{t+1}) - g_t(\mathbf{L}_t)]^-|\mathcal{F}_t] \\ &\quad< \infty.\end{aligned} \tag{56}$$

As demonstrated in Propositions B.2 and B.4, $g_t(\mathbf{L})$ and $f_t(\mathbf{L})$ are both Lipschitz with respect to $\mathbf{L}$, which implies there exists κ_2 such that

$$\begin{aligned}&|g_{t+1}(\mathbf{L}_{t+1}) - f_{t+1}(\mathbf{L}_{t+1}) - (g_t(\mathbf{L}_t) - f_t(\mathbf{L}_t))| \\ &\quad= |g_{t+1}(\mathbf{L}_{t+1}) - g_t(\mathbf{L}_t) - (f_{t+1}(\mathbf{L}_{t+1}) - f_t(\mathbf{L}_t))| \\ &\quad\le |g_{t+1}(\mathbf{L}_{t+1}) - g_t(\mathbf{L}_t)| + |f_{t+1}(\mathbf{L}_{t+1}) - f_t(\mathbf{L}_t)| \\ &\quad\le |g_{t+1}(\mathbf{L}_{t+1}) - g_{t+1}(\mathbf{L}_t)| + |g_{t+1}(\mathbf{L}_t) - g_t(\mathbf{L}_t)| \\ &\qquad + |f_{t+1}(\mathbf{L}_{t+1}) - f_{t+1}(\mathbf{L}_t)| + |f_{t+1}(\mathbf{L}_t) - f_t(\mathbf{L}_t)| \\ &\quad\le \kappa_2\|\mathbf{L}_{t+1} - \mathbf{L}_t\|_F.\end{aligned} \tag{57}$$

According to Lemma A.5, combining with $\sum_{t=1}^{\infty}(1/(t+1)) = \infty$, we have that

$$\lim_{t\to\infty} g_t(\mathbf{L}_t) - f_t(\mathbf{L}_t) = 0, a.s.. \tag{58}$$

■

Theorem C.4: *As t tends to infinity, given the $\mathbf{L}_t$ is obtained by Algorithm 1, $g_t(\mathbf{L}_t) - f(\mathbf{L}_t)$ converges to 0 almost surely.*

Proof: From Theorem C.2, we have $\|f - f_t\|_\infty$ converging to 0 almost surely, which implies that

$$\lim_{t\to\infty} g_t(\mathbf{L}_t) - f(\mathbf{L}_t) = 0, a.s.. \tag{59}$$

■

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Junpeng Zhang (Student Member, IEEE) received the B.Sci. degree and the master's degree in surveying engineering from the China University of Mining and Technology, Xuzhou, China, in 2013 and 2016, respectively. He is currently pursuing the Ph.D. degree in electrical engineering with the University of New South Wales, Canberra, ACT, Australia.

His research interests include object detection and tracking in remote sensing imaginary.

Mr. Zhang was the winner of the "DSTG Best Contribution to Science Award" in Digital Image Computing: Techniques and Applications 2018 (DICTA 2018). He is serving as the Chair for the IEEE GRSS UNSW Canberra Student Chapter.



Xiuping Jia (Senior Member, IEEE) received the B.Eng. degree from the Beijing University of Posts and Telecommunications, Beijing, China, in 1982, and the Ph.D. degree in electrical engineering from the University of New South Wales, Canberra, ACT, Australia, in 1996.

Since 1988, she has been with the School of Engineering and Information Technology, University of New South Wales at Canberra, where she is currently an Associate Professor. She has authored or coauthored more than 200 referred articles, including over 100 journal articles. She has coauthored the remote sensing textbook titled *Remote Sensing Digital Image Analysis* [Springer-Verlag, Third (1999) and Fourth Editions (2006)]. Her research interests include remote sensing, image processing, and spatial data analysis.

Dr. Jia is a Subject Editor of the *Journal of Soils and Sediments* and an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Jiankun Hu (Senior Member, IEEE) received the Ph.D. degree in control engineering from the Harbin Institute of Technology, Harbin, China, in 1993, and the master's degree in computer science and software engineering from Monash University, Melbourne, VIC, Australia, in 2000.

He was a Research Fellow with Delft University, Delft, The Netherlands, from 1997 to 1998, and The University of Melbourne, Melbourne, from 1998 to 1999. He is a Full Professor of cybersecurity with the School of Engineering and Information Technology, University of New South Wales at Canberra, Canberra, ACT, Australia. His main research interest is in the field of cybersecurity, including biometrics security, where he has published many articles in high-quality conferences and journals, including the IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE.

Dr. Hu has served on the editorial boards of up to seven international journals and served as the Security Symposium Chair for the IEEE Flagship Conferences of the IEEE International Conference on Communications (ICC) and the IEEE Global Communications Conference (GLOBECOM). He has obtained nine Australian Research Council (ARC) Grants. He served at the prestigious Panel of Mathematics, Information and Computing Sciences, ARC Excellence in Research for Australia (ERA) Evaluation Committee 2012.



Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from the Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

Since 1999, he has been with Grenoble INP, where he is currently a Professor of signal and image processing. He has been a Visiting Scholar with Stanford University, Stanford, CA, USA, KTH, Stockholm, Sweden, and NUS, Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavik, Iceland. From 2015 to 2017, he was a Visiting Professor with the University of California at Los Angeles (UCLA), Los Angeles, CA. He holds the AXA Chair in remote sensing and is an Adjunct Professor with the Chinese Academy of Sciences, Aerospace Information Research Institute, Beijing. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot is the founding President of the IEEE Geoscience and Remote Sensing French Chapter from 2007 to 2010, which received the 2010 IEEE GRSS Chapter Excellence Award. He was a member of the Institut Universitaire de France from 2012 to 2017 and a Highly Cited Researcher of Clarivate Analytics/Thomson Reuters from 2018 to 2019. He was the Vice-President of the IEEE Geoscience and Remote Sensing Society, where he was in charge of meetings and symposia from 2017 to 2019. He has received multiple outstanding paper awards. He was the General Chair of the first IEEE GRSS Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote sensing (WHISPERS). He was the Chair from 2009 to 2011 and the Co-Chair of the GRS Data Fusion Technical Committee from 2005 to 2008. He was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008 and the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON IMAGE PROCESSING, and the PROCEEDINGS OF THE IEEE. He was the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. In 2014, he served as a Guest Editor for the *IEEE Signal Processing Magazine*.

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